

TITLE: Responsible AI and Model Interpretation

1. Objective

The objective of this task is to interpret the machine learning model and understand which features influence predictions the most.

2. Model Used

Random Forest Classifier

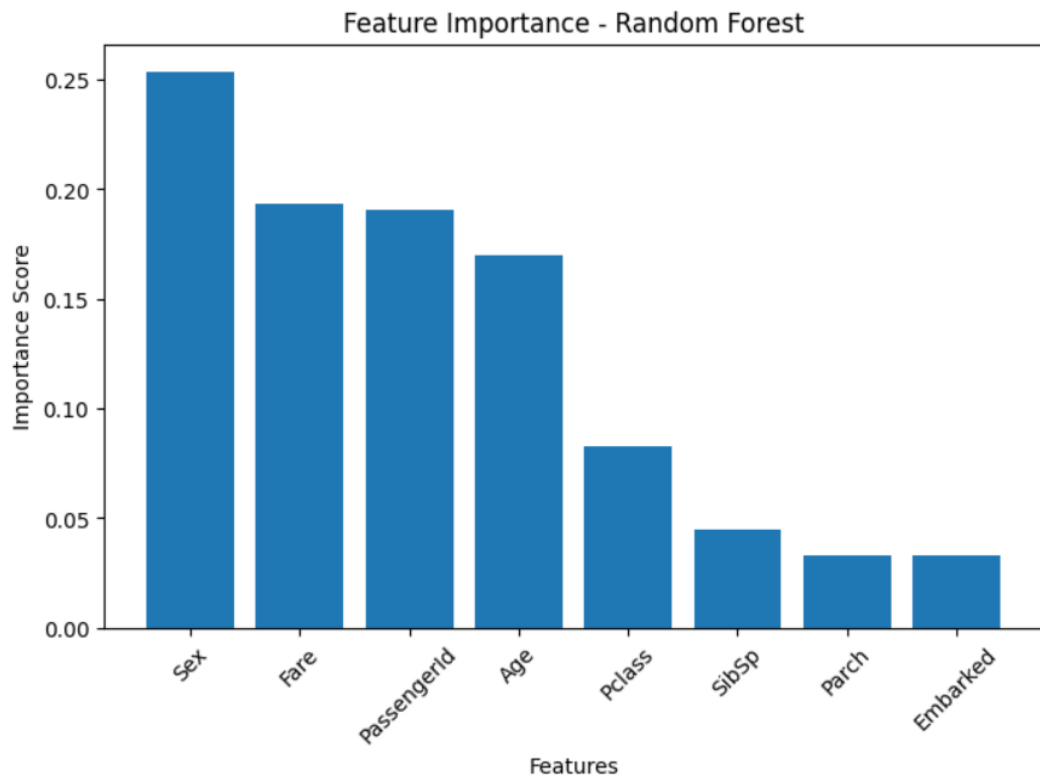
3. Feature Importance Analysis

Feature importance scores were calculated using the Random Forest model.

Top Important Features:

- Sex
- Fare
- PassengerId
- Age
- Pclass

4. Feature Importance Visualization



5. Bias Analysis

The model relies heavily on features such as Sex and Fare. This may introduce bias because different passenger groups had different survival rates.

6. Mitigation Recommendations

- Use balanced datasets
- Remove unnecessary features
- Regularly monitor model fairness
- Evaluate performance across different groups

7. Conclusion

Feature importance analysis helped explain how the model makes predictions. Responsible AI practices improve transparency and trust in machine learning systems.